

§10. Syzygies

Def. F minimal graded free resolution of U . $\cdots \rightarrow F_2 \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \xrightarrow{d_0} U$

$$\text{Syz}_i^R(U) := \text{Im}(d_i) = \ker(d_{i-1}) \cong (\ker(d_{i+1})) \quad i \geq 1$$

$$\text{Syz}_0^R(U) := U$$

$$\begin{array}{ccccccc} & & & 0 & & & 0 \\ & & & \searrow & & \nearrow & \\ & & & \text{Syz}_i^R(U) & & & \\ & & & \downarrow & & \uparrow & \\ \cdots \rightarrow F_{i+2} & \rightarrow & F_{i+1} & \xrightarrow{d_{i+1}} & F_i & \xrightarrow{d_i} & F_{i-1} \rightarrow \cdots \\ & & \downarrow & \nearrow & \downarrow & \nearrow & \\ & & \text{Syz}_{i+1}^R(U) & & \text{Syz}_i^R(U) & & \\ & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & \end{array}$$

§11 & §12.

Def. Betti number of U

$$b_i^R(U) := \text{rank}(F_i)$$

projective dimension of U

$$\text{pdr}(U) = \max\{i \mid b_i^R(U) \neq 0\}$$

Poincaré series of U

$$P_U^R(t) = \sum_{i \geq 0} b_i^R(U) t^i$$

Graded betti number

$$b_{i,p}^R(U) := \text{number of summands in } F_i \text{ of the form } R(-p)$$

Graded Poincaré series

$$P_U^R(t, z) = \sum_{i \geq 0, p \in \mathbb{Z}} b_{i,p}^R(U) t^i z^p$$

$$(P_U^R(t) = P_U^R(t, 1))$$

Thm 11.2

$$b_i^R(U) = \dim_k(\text{Tor}_i^R(U, k))$$

$$= \dim_k(\text{Ext}_R^i(U, k))$$

pf. For Tor

$$F \otimes_R k \rightarrow F_i \otimes_R k = R^{b_i^R(U)} \otimes_R k = k^{b_i^R(U)}$$

Since $\text{Im}(d_i) \subseteq m F_i$

$$F \otimes_R k = \cdots \rightarrow k^{b_i^R(U)} \xrightarrow{0} k^{b_{i-1}^R(U)} \xrightarrow{0} \cdots$$

$$\Rightarrow \text{Tor}_i^R(U, k) \cong H_i(F \otimes_R k) = k^{b_i^R(U)}$$

□

Ex 12.2

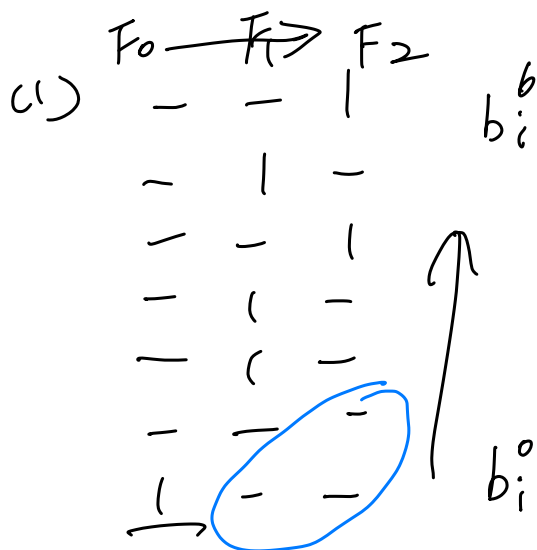
$$A = k[x, y], \quad B = (x^3, xy, y^5) \quad A/B$$

$$0 \rightarrow A(-4) \oplus A(-6) \rightarrow A(-3) \oplus A(-2) \oplus A(-5)$$

$$\rightarrow A \rightarrow A/B \rightarrow 0$$

$$\uparrow \quad \uparrow$$

$$F_0 \quad F_0$$



(2)

	1	2	3
0	1	-	-
1	-	1	-
2	-	1	1
3	-	-	-
4	-	-	1

prop 12.3. $\underline{c}$ is the minimal degree of an element in a mini system of homo generators of U . We have $b_{i,p}^R(U) = 0$ for $p < i + c$.

pf. Induction on i .

(1) $i = 0$, $F_0 \xrightarrow{F_1} U \rightarrow 0$
 $p < c \Rightarrow b_{0,p}^R(U) = 0$

(2) Suppose it holds for i .

If F_i is minimal $\Rightarrow \text{im}(d_{i+1}) \subseteq mF_i$.

If $\exists p < i + 1 + c$ s.t. $b_{i+1,p}^R(U) \neq 0$,
 then $(mF_i)_p \neq 0$

$\Rightarrow \exists p_0 < i + c$ s.t. $(F_i)_{p_0} \neq 0$

□

§13. The connecting homomorphism

$$0 \rightarrow (F, d) \rightarrow (F', d') \rightarrow (F'', d'') \rightarrow 0 \text{ exact}$$

$$\begin{aligned}
 \Rightarrow \dots \rightarrow H_i(F) \rightarrow H_i(F') \rightarrow H_i(F'') \rightarrow H_{i-1}(F) \\
 \rightarrow \dots \\
 \text{exact}
 \end{aligned}$$

Thm 13.4 $0 \rightarrow U \rightarrow U' \rightarrow U'' \rightarrow 0$ exact

F, F'' graded free reso of U, U''

$\Rightarrow \exists$ a graded free reso F' of U'

$$0 \rightarrow F \rightarrow F' \rightarrow F'' \rightarrow 0 \text{ split exact}$$

pf. Set $F_i' = F_i \oplus F_i''$

$$0 \rightarrow F_i \xrightarrow{\pi} F_i' \xrightarrow{\pi} F_i'' \rightarrow 0$$

✓

□